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RESEARCH ARTICLE

Determinants of global variation in taxonomic and phylogenetic diversity of invasive plants

Bi-Cheng Dong[1,2] | Lan-Hui Wang[3] | Li-Yuan Gao[3] | Mark van Kleunen[4,5,6] | Fei-Hai Yu[1,5,7]

[1]School of Ecology and Nature Conservation, Beijing Forestry University, Beijing, China; [2]The Key Laboratory of Ecological Protection in the Yellow River Basin of National Forestry and Grassland Administration, Beijing Forestry University, Beijing, China; [3]School of Economics and Management, Beijing Forestry University, Beijing, China; [4]Ecology, Department of Biology, University of Konstanz, Konstanz, Germany; [5]Zhejiang Provincial Key Laboratory of Plant Evolutionary Ecology and Conservation, Taizhou University, Taizhou, Zhejiang, China; [6]Zhejiang Key Laboratory for Restoration of Damaged Coastal Ecosystems, School of Life Sciences, Taizhou University, Taizhou, Zhejiang, China and [7]School of Life and Environmental Sciences, Shaoxing University, Shaoxing, Zhejiang, China

Correspondence
Fei-Hai Yu
Email: feihaiyu@126.com

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Abstract

1. Information on the determinants of taxonomic and phylogenetic diversity of invasive plant species is crucial for managing invasive plants. With globalization, most countries have experienced substantial economic losses and environmental damage due to biological invasions. We analysed the determinants of variation in the diversity and phylogenetic structure of invasive plants among countries worldwide.
2. To do so, we used a comprehensive checklist of invasive plants in 152 countries worldwide to calculate taxonomic and phylogenetic diversity (i.e. Faith's PD metric) and phylogenetic structure, using mean phylogenetic distance (MPD) and mean nearest taxon distance (MNTD). We then combined these data in minimum adequate models with data on geographic, climatic, socio-economic and international trade variables. We also conducted randomization tests to determine whether the phylogenetic diversity of invasive plants in these countries was clustered or overdispersed.
3. Taxonomic and phylogenetic diversity of invasive plants exhibited spatial congruence. Taxonomic and phylogenetic diversity were positively correlated with insularity, mean annual precipitation (MAP) and HS-12 (seeds, grains and medicinal plants) import values per capita, but negatively associated with mean annual temperature (MAT) and HS-07 (vegetables) import evenness. In addition, taxonomic diversity also increased with airport density, HS-12 import evenness and lower HS-08 (fruit and nuts) imports. MPD increased with greater land area and airport density and fewer HS-12 exporting source countries. MNTD increased with MAT but declined with greater land area and insularity. Phylogenetic clustering

Bi-Cheng Dong and Lan-Hui Wang contributed equally.

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occurred in 28.9%–49.3% of countries, whereas phylogenetic overdispersion was rare, observed only in 0.6%–5.3% of countries.

4. *Synthesis*. Our study reveals that variation in taxonomic and phylogenetic diversity of invasive plant species among countries is shaped by geographic, socio-economic, climatic and international trade factors. Nearly one-third of the countries showed phylogenetic clustering of invasive plant species, indicating a relatively consistent global pattern. These findings underscore the importance of integrating both taxonomic and phylogenetic perspectives in invasion ecology, emphasizing the need for regionally tailored management strategies that effectively account for regional geographic, climatic, socio-economic and trade-related factors to mitigate future plant invasions.

KEYWORDS
geographic isolation, international trade, phylogenetic clustering, plant invasions, regional climates, socio-economic characteristics, taxonomic and phylogenetic diversity

1 | INTRODUCTION

Globalization has become a major force in the world, shaping the economy, society and biotas worldwide (Hulme, 2021). While globalization has accelerated the movement of goods across geographical boundaries, it has also facilitated the spread of alien species across biogeographical barriers (Capinha et al., 2023; Faulkner et al., 2020; Sardain et al., 2019; Seebens et al., 2015). As of 2020, around 3000 vascular plant species have already been documented as invasive outside their native ranges (Laginhas et al., 2023; Pagad et al., 2022), and the number of invasive plant species is still increasing and is far from reaching saturation (Seebens et al., 2015, 2021). Understanding the current taxonomic and phylogenetic diversity patterns of invasive plant species and their determinants is crucial for developing effective measures and regulations to prevent further invasions.

International trade, an inherent aspect of globalization, can increase the chances of introducing invasive plants (Egawa & Koyama, 2023; Ollier & Bertelsmeier, 2022; Seebens et al., 2015). Many plants with economic uses, such as crops or ornamental plants, have been intentionally introduced for agricultural or horticultural purposes (Guo et al., 2024; Mokotjomela et al., 2023; van Kleunen et al., 2018, 2020). These introduced economic plants are often actively propagated and repeatedly planted in large numbers over large areas, thereby having a high likelihood of establishment (Dong et al., 2024; Guo et al., 2024; Pironon et al., 2024; van Kleunen et al., 2020). The movement of goods and commodities across borders provides opportunities for unintentionally introducing invasive plants. This can occur through the transport of plant materials, such as grains or plant cuttings, which are often contaminated with viable propagules of invasive plants (Ikeda et al., 2022; Yu et al., 2020). Most studies that assessed the invasion risk of alien plants through international trade used cumulative imports of general commodities (e.g. Levine & D'Antonio, 2003). However, there is still a lack of knowledge regarding whether the global distribution of invasive plants is linked to the imports of specific commodities worldwide (see also Bradley et al., 2012; Fenn-Moltu et al., 2023; Ollier & Bertelsmeier, 2022).

Aside from the imports of commodities via international trade, the introduction of invasive plants across borders is influenced by the geographic and socio-economic characteristics of recipient regions (Essl et al., 2019; Guo et al., 2024; Pyšek et al., 2017; Seebens et al., 2015). First, human pressures, as estimated with parameters such as the gross domestic product (GDP) per capita, human population size and the density of port infrastructure (e.g. airports and seaports), can strongly influence the likelihood of successful establishment of invasive plants in a specific country. It is reasonable to assume that countries with higher GDP per capita, larger human population sizes and higher densities of port infrastructure may be more likely to introduce a greater number of alien plants into a country, consequently leading to a higher prevalence of invasive plants (Dawson et al., 2017; Faulkner et al., 2020). Second, climatic variation among recipient countries—particularly with regard to temperature and precipitation—may also influence the variation in establishment success of invasive plants. This may be true when the native floras of countries are not equally susceptible to invasion by newly arriving alien plants or expansion by already established invasive plants (Finch et al., 2021; Walther et al., 2009). Furthermore, it is essential to consider the geographic features, including the size of the recipient country and whether it is part of the mainland or is an island or archipelago, which potentially influences the likelihood that these plants become invasive (Essl et al., 2019; Schrader et al., 2021).

Besides taxonomic diversity (i.e. the number of invasive species), globalization may also influence the phylogenetic diversity and structure of invasive plants in recipient countries. Because international trade via globalization offers an opportunity to introduce goods from a variety of sources into a particular region, this may, to some extent, contribute to a higher likelihood of introducing invasive plants from different phylogenetic lineages (e.g. Yu et al., 2020).

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Therefore, it is reasonable to assume that globalization tends to result in the introduction of plants from distantly related alien floras, which may increase phylogenetic diversity among invasive plants in recipient countries (Parker et al., 2012; Zhang et al., 2021). To date, however, most studies have focused on globalization's effects on the taxonomic diversity of invasive plants (e.g. Essl et al., 2019; Humair et al., 2015), and little is known about the determinants of phylogenetic diversity and structure.

To increase our knowledge about drivers of global invasion success, we used a comprehensive dataset of 2979 invasive plant species in 152 countries worldwide. We performed minimum adequate models to test whether taxonomic diversity and phylogenetic diversity of invasive plants are associated with the international trade of recipient countries and their geographic, climatic and socio-economic features. In addition, investigating the phylogenetic structure of invasive plants is crucial to understanding the mechanisms driving plant invasions, as it provides insights into their evolutionary history and local adaptability. We, therefore, also tested whether the phylogenetic diversity of invasive plants is clustered or overdispersed in the countries. In particular, we hypothesize that (1) taxonomic and phylogenetic diversity of invasive plants in countries are positively associated with the imports of specific commodities. (2) Both types of diversity are positively associated with higher human pressures as measured by GDP per capita, human population size and port infrastructure density. (3) Both diversity are positively associated with country-specific geographic characteristics such as land area and insularity (with islands demonstrating a higher vulnerability than mainland regions) and are modulated by climate variables (temperature and precipitation). (4) Developed countries with established trade networks exhibit phylogenetic clustering of invasive plants through repeated introductions of related species, while developing regions with less systematic trade relationships exhibit phylogenetic overdispersion, reflecting diverse introduction pathways.

2 | METHODS

2.1 | Data on invasive species

We first extracted the global checklist of invasive plant species from the dataset of *the Global Register of Introduced and Invasive Species* (GRIIS) (Pagad et al., 2022; version 1_0, https://zenodo.org/records/6348164). Because the original species names in GRIIS include many with taxonomic uncertainty (e.g. absence of author names, lexical variants, synonyms, intraspecific names and potentially different taxonomic interpretations), we followed the instructions of Matilda Brown (https://matildabrown.github.io/rWCVP/articles/redlist-name-matching.html) and harmonized these species names according to the *World Checklist of Vascular Plants* (WCVP; version 12), using the R package 'rWCVP' (Brown et al., 2023). We then retained the invasive species distributed exclusively in terrestrial and/or freshwater habitats, and we only included countries with more than one invasive species. This resulted in 2979 invasive plant species with accepted names, occupying 152 countries.

2.2 | Measures of taxonomic and phylogenetic diversity

Taxonomic diversity was measured as species richness. Phylogenetic diversity was measured with Faith's PD metric, that is, the sum of branch lengths in the phylogenetic assemblage. Furthermore, two complementary metrics were used to analyse phylogenetic structure, including mean pairwise distance (MPD) and mean nearest taxon distance (MNTD). MPD calculates the average phylogenetic distance of all possible pairs of invasive species in a country, while MNTD measures the mean phylogenetic distance to the closest invasive relative for each invasive species. These two metrics provide different evolutionary insights, that is, MPD is sensitive to large phylogenetic distances contributed by co-occurring species belonging to old and distinct lineages, whereas MNTD is influenced by more recent speciation only (Eme et al., 2020). Phylogenetic diversity, MPD and MNTD were calculated using the R package 'Picante' (Kembel et al., 2010).

To construct a phylogenetic tree of the 2979 invasive species and obtain the relevant phylogenetic information, we matched their species names to the tips of a megatree from Smith and Brown (2018) and Jin and Qian (2019), using the R package 'rtrees' (Li, 2023). Species not present in the megatree were added as polytomies at the basal nodes of their respective genera (22.6% of total species) or, if unavailable, at the family level (5.1% of total).

2.3 | Factors potentially influencing invasive plant diversity

To identify factors that potentially influence the taxonomic and phylogenetic diversity of invasive plants worldwide, we first compiled socio-economic variables for each country, including GDP, human population density and land area. GDP was estimated by averaging values from 2016 to 2020 for each country, using data from the *CEPII Gravity Database* (Conte et al., 2022). The human population density was estimated by averaging values for the same period based on the *World Population Prospects 2022* provided by the United Nations Department of Economic and Social Affairs (https://population.un.org/wpp). Land area (expressed in square kilometres, km^2) was estimated using total population and population density data from *the World Population Prospects 2022*. Since country-wise GDP typically correlates with large populations, we calculated GDP per capita (expressed in thousand US dollars per person).

In addition, we assessed the role of insularity in plant invasions by classifying each country as an island (including archipelagos) or as part of the mainland. Information on each country's insularity

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was obtained from *Wikimili* (https://wikimili.com/en/List_of_island_countries) and other relevant online sources.

We compiled international trade data from the *UN Comtrade Database* (https://comtradeplus.un.org/; accessed on 3 September 2024). This database categorizes goods using the *Harmonized Commodity Description and Coding System* (HS), a hierarchical system comprising six-digit codes. The first two digits (HS-2) classify products into specific chapters (World Customs Organization, 2023). We collected trade data for 152 importing countries (i.e. the countries with data on invasive species) and 231 exporting countries across four levels of HS-2 codes related to imports of live plant products, including HS-06 (live plants and cut flowers), HS-07 (vegetables), HS-08 (fruit and nuts) and HS-12 (seeds, grains and medicinal plants) (Table S1) for the period from 2016 to 2020. To mitigate the potential effects of inflation on GDP and HS-2 commodity data, we selected 2016 as the base year. We adjusted all values to 2016 dollars by multiplying each year's nominal value by the ratio of the Consumer Price Index (CPI) of 2016 to that specific year's CPI. This conversion formula (Real value = Nominal value × $CPI_{2016}/CPI_{specific\ year}$) standardized all monetary values to 2016 purchasing power. These trade variables were then calculated per capita (expressed in US dollars per person). Moreover, for each importing country and each HS category, we characterized their import source diversity through two measures: (1) the number of source countries that exported each HS category and (2) the evenness in import values of each HS category across source countries. The evenness was calculated using Shannon's evenness index (J'), where $J' = H'/\ln(S)$. H' is Shannon's diversity index calculated as $H' = -\Sigma(p_i \times \ln(p_i))$, where p_i is the proportion of import values from a source country i to total import values within each HS category, and S is the number of source countries, following the concept of species evenness in ecological communities (Magurran, 2004).

Additionally, we used port infrastructures as proxies for international trade by extracting information on airports from the *Global Airport Database* (http://www.partow.net/miscellaneous/airportdatabase/index.html), and on seaports from the *Maritime Safety Information* (https://msi.nga.mil/Publications/WPI) for each country. Since larger countries typically have more airports and seaports, we calculated airport density and seaport density (expressed in number per 100,000 km^2).

We evaluated the effects of climatic variables by calculating mean annual temperature (MAT; °C) and mean annual precipitation (MAP; mm) per country from the *WorldClim* dataset (version 2.1; https://worldclim.org) at a 10′ × 10′ resolution. These two climatic variables were selected because they are known to have strong influences on plant species distributions (Root et al., 2003) and plant phylogenetic structure (Qian et al., 2022).

2.4 | Data analysis

To achieve more regular distributions of the skewed continuous predictor variables, we natural-log(x + 1)-transformed the density of airports and seaports, natural-log(x + 10)-transformed MAT and natural-log(x)-transformed other continuous predictor variables (except the evenness in import values of each HS category). To facilitate comparisons between the effects of different predictor variables, we also scaled the transformed variables to have a mean of zero and a standard deviation of one (Schielzeth, 2010). To achieve more regular distributions of response variables, we natural-log(x)-transformed phylogenetic diversity, MPD and MNTD.

To evaluate the taxonomic diversity of invasive plants (i.e. count data), we started the analysis by building a negative binomial generalized linear model (negative binomial GLM) and all other predictor variables as fixed factors, using the 'glm.nb' function in the R package 'MASS' (Venables & Ripley, 2002). The scaled logarithm of the land area was included as an offset in the model (Zuur et al., 2009), because both invasive and native plant species diversity are area-dependent, that is, larger countries may harbour more native and invasive species and greater phylogenetic diversity overall (Figures S3 and S4). To evaluate the phylogenetic diversity, MPD and MNTD of invasive plants (i.e. continuous data), we started the analysis by building linear models (LMs) with all predictor variables as fixed factors. To obtain minimum adequate models, we performed variable selection using a backward stepwise procedure based on the exact Akaike's information criterion (AIC), with the 'step' function implemented in the R package 'stats'. To mitigate the multicollinearity problem, variables with a variance inflation factor (VIF) greater than 5 were eliminated using the 'stepVIF' function implemented in the R package 'pedometrics' (Samuel-Rosa, 2022). The variable selection procedure began with a full model and automatically ceased when the optimized model had the lowest AIC and all remaining variables had VIF values below 5. After this, we manually excluded any non-significant terms from the optimal model. Finally, we employed hierarchical partitioning methods to calculate individual marginal R^2 values for each fixed factor to quantify the proportional contribution of each factor within the LMs, using the 'glmm.hp' package (Lai et al., 2022, 2023).

To determine whether the phylogenetic diversity of invasive plants in each country is more clustered or overdispersed than expected by chance, we employed a randomization test. First, we created potential country-specific invasive species pools for each country. This was done using the global invasive species pool of 2979 species and excluding from the ones that are native to the respective country. Data on the latter were obtained from the *WCVP* (version 12; https://sftp.kew.org/pub/data-repositories/WCVP/Archive/). To assess phylogenetic diversity, MPD and MNTD, we conducted 999 randomizations while maintaining the observed invasive species richness for each country by randomly selecting species from each country-specific potential invasive species pool. We then compared the phylogenetic diversity, MPD and MNTD of invasive species in each country to the expected values based on the 999 randomizations. If the observed value fell below the 2.5th percentile or above the 97.5th percentile of the resampled values, we considered it significantly lower or higher than expected, respectively. This approach allowed us to identify phylogenetic clustering

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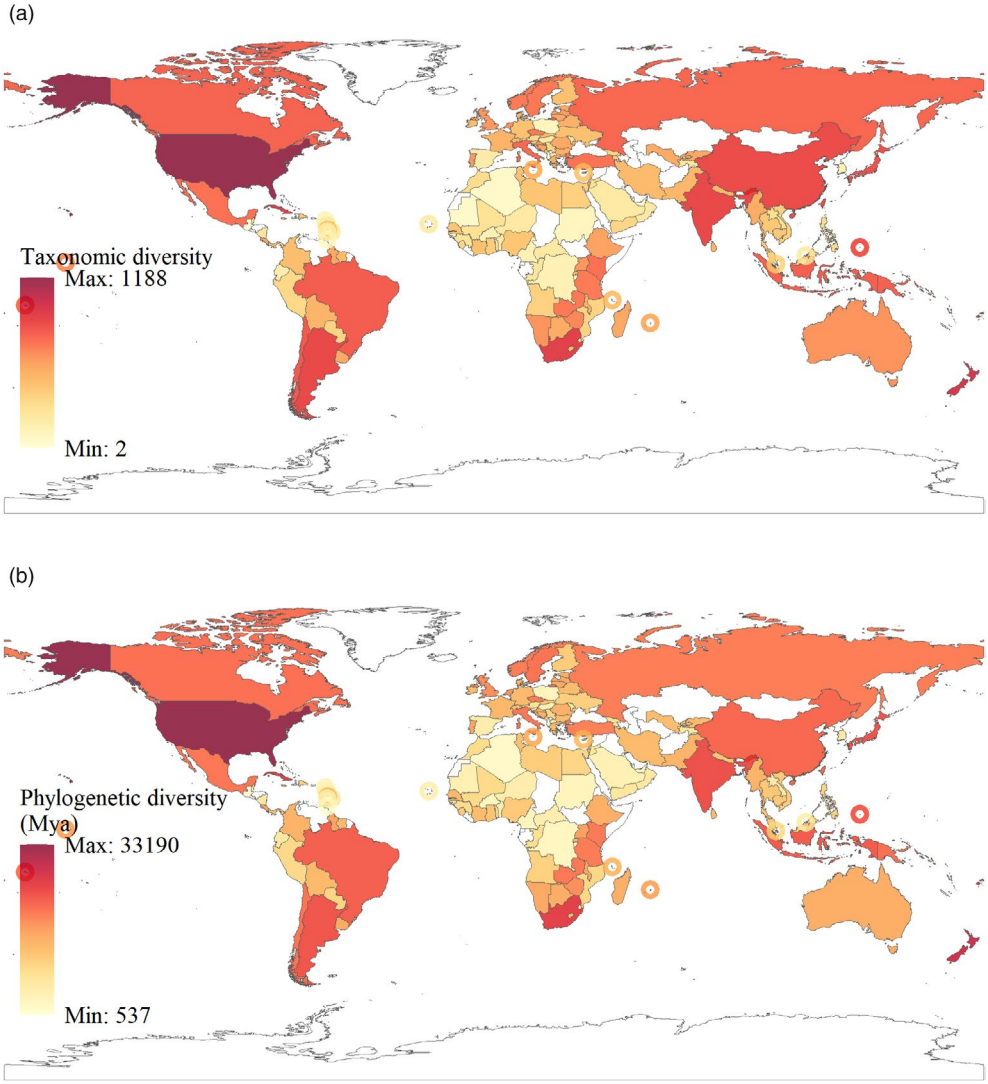

FIGURE 1 Global distribution of taxonomic diversity (a) and phylogenetic diversity (b) of invasive plants. Data on taxonomic diversity and phylogenetic diversity are available for 152 countries with more than one invasive species. Island countries smaller than 10,000 km^2 ($n = 16$) are marked with circles to ensure their visibility on the map. Regions with white filling have no data.

(i.e. observed values lower than expected) or phylogenetic overdispersion (i.e. observed values higher than expected) among invasive species in each country.

3 | RESULTS

3.1 | Determinants of taxonomic diversity

The GRIIS database provided a comprehensive overview of invasive plant species across 152 countries worldwide, comprising 33 island countries and 119 mainland countries. The number of invasive plants varied significantly among countries, with a median of 28 species per country. Among mainland countries, the United States has the highest number of alien plant species recorded in GRIIS, with 1188 species. Among island countries, New Zealand leads with 426 invasive plant species recorded in GRIIS (Figure 1a).

The minimum adequate model of taxonomic diversity, which accounts for differences in the area of the countries, included eight of the 19 explanatory variables (Table 1a). Taxonomic diversity increased significantly with increasing airport density, MAP, import values of the HS-12 category per capita and the evenness in import values of the HS-12 category, and it was higher when the country was an island or archipelago (Figure 2a,b,d,f,h). In contrast, taxonomic diversity decreased significantly with increasing MAT, import values of the HS-08 category per capita and the evenness in import values of the HS-07 category (Figure 2c,e,g).

3.2 | Determinants of phylogenetic diversity and structure

The phylogenetic diversity of invasive plants showed a geographic distribution similar to the one for taxonomic diversity (Figure 1b).

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TABLE 1 Results of the minimum adequate models testing the effects of explanatory variables on the (a) taxonomic and (b) phylogenetic diversity of invasive plants in 152 countries worldwide.

Term	Estimate	SE	*t*-value	*p*-value	Ind. R^2	Ind. perc.
(a) Taxonomic diversity						
Intercept	3.48	0.09	39.09	<0.001	/	/
Insularity	1.39	0.23	5.99	<0.001	/	/
Airport density	0.32	0.1	3.11	0.002	/	/
MAT	−0.32	0.09	−3.47	0.001	/	/
MAP	0.26	0.09	2.99	0.003	/	/
HS-08 per capita	−0.31	0.16	−2.00	0.046	/	/
HS-12 per capita	0.52	0.15	3.42	0.001	/	/
HS-07: evenness	−0.27	0.09	−3.06	0.002	/	/
HS-12: evenness	0.24	0.09	2.64	0.008	/	/
(b) Phylogenetic diversity						
Intercept	7.65	0.06	119.08	<0.001	/	/
Insularity	0.74	0.17	4.44	<0.001	0.055	16.57
Area	0.45	0.07	6.63	<0.001	0.128	38.33
MAT	−0.14	0.06	−2.18	0.031	0.039	11.69
MAP	0.19	0.06	3.18	0.002	0.037	10.97
HS-12 per capita	0.20	0.06	3.13	0.002	0.040	11.99
HS-07: evenness	−0.16	0.06	−2.83	0.005	0.035	10.46

Note: The model of taxonomic diversity employed a negative binomial generalized linear model with land area as an offset variable, while the model of phylogenetic diversity utilized a linear model. The marginal R^2 of the model is 0.55 for taxonomic diversity and 0.33 for phylogenetic diversity. Individual R^2 values for each fixed factor and their percentage contribution to the overall R^2 are given only for the linear model of phylogenetic diversity. Trade categories retained in the models are HS-07 (vegetables), HS-08 (fruit and nuts) and HS-12 (seeds, grains and medicinal plants).

The minimum adequate model of phylogenetic diversity included six of the 20 explanatory variables (Table 1b). Phylogenetic diversity increased significantly with increasing land area, MAP and import values of the HS-12 category per capita, and it was higher when the country was an island or archipelago (Figure 3a,b,d,e). In contrast, phylogenetic diversity decreased significantly with increasing MAT and the evenness in import values of the HS-07 category (Figure 3c,f). Land area accounted for 38.33% of the variation in phylogenetic diversity, followed by insularity, import value of the HS-12 category per capita, MAT, MAP and evenness in import values of the HS-07 category (Table 1b).

For phylogenetic structure, MPD increased significantly with increasing land area and airport density and decreased with the number of source countries that exported the HS-12 category (Figures 4a and 5a–c). The number of source countries that exported the HS-12 category accounted for 43.99% of the variation in MPD, followed by land area and airport density (Table 2a). Furthermore, MNTD increased significantly with increasing MAT but decreased with increasing land area (Figures 4b and 5d,f). MNTD was also significantly lower when the country was an island or archipelago (Figure 5e). MAT accounted for 59.41% of the variation in MNTD, followed by land area and insularity (Table 2b).

3.3 | Regional clustering of phylogenetic diversity

For phylogenetic diversity, the randomization test revealed three distinct patterns of phylogenetic diversity. Forty-seven countries (30.9%) exhibited significantly lower phylogenetic diversity than expected, indicating strong phylogenetic clustering (Figure 6a). Notable examples included China, Chile and Russia (Table S2). Mauritania emerged as the sole country with a significantly higher phylogenetic diversity than expected, suggesting phylogenetic overdispersion (Figure 6a). The remaining 104 countries (68.4%) showed phylogenetic diversity patterns within the range of expected values (Figure 6a).

For MPD, 44 countries (28.9%) demonstrated significantly lower MPD than expected, indicating strong phylogenetic clustering (Figure 6b). Notable examples included India, China and Russia (Table S2). In contrast, eight countries (5.3%) exhibited significantly higher MPD than expected ($p < 0.025$), suggesting

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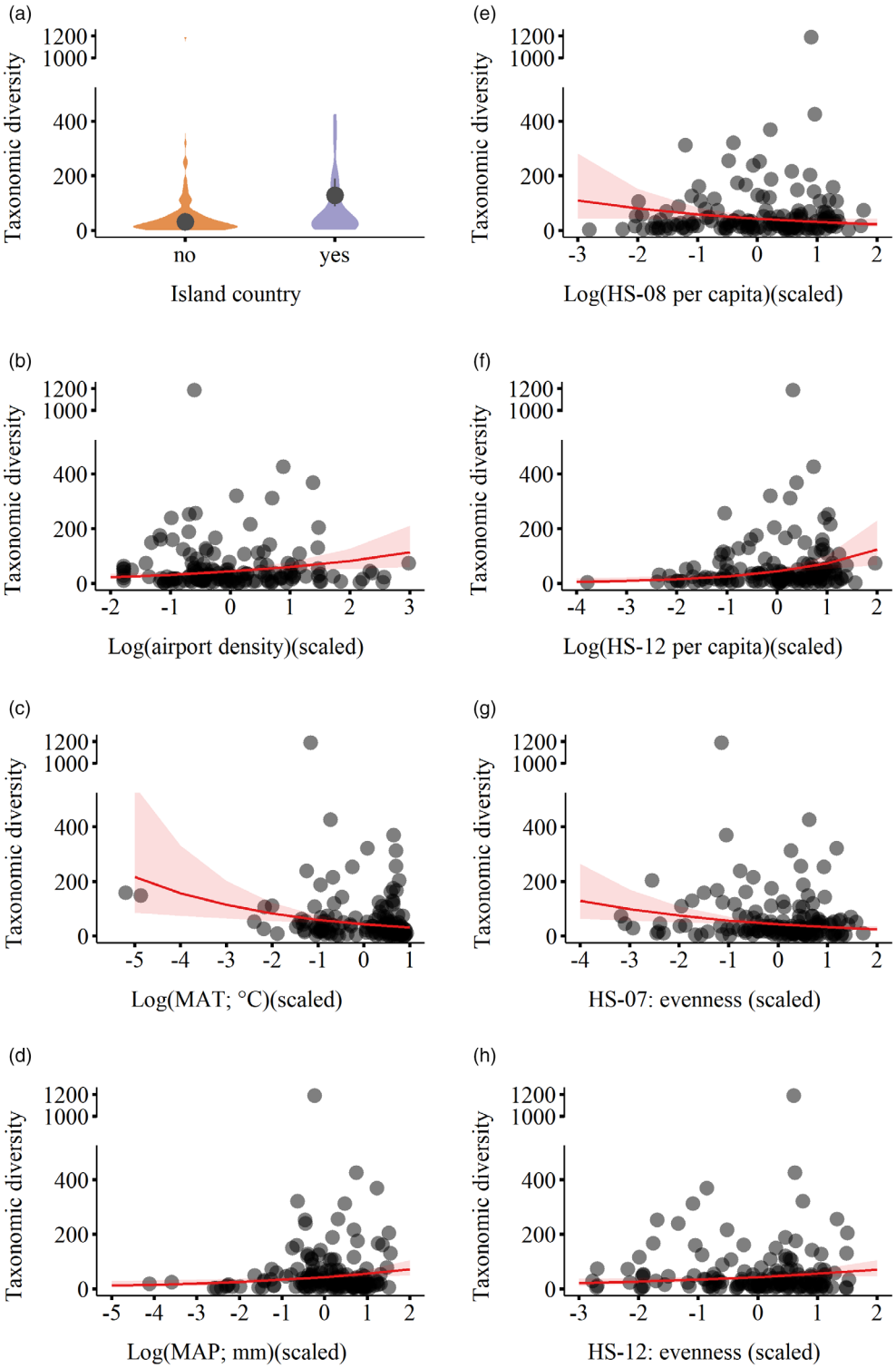

FIGURE 2 Taxonomic diversity of invasive plants in 152 countries around the world, in relation to (a) insularity, (b) airport density, (c) mean annual temperature (MAT), (d) mean annual precipitation (MAP), (e) import values of the HS-08 category (fruit and nuts) per capita, (f) import values of the HS-12 category per capita (seeds, grains and medicinal plants), (g) evenness in import values of the HS-07 category (vegetables) and (h) evenness in import values of the HS-12 category. The fitted lines with 95% confidence intervals from the minimum adequate negative binomial GLM model, with the log of country area as an offset, are shown.

phylogenetic overdispersion (Figure 6b). The remaining 100 countries (65.8%) exhibited MPD values within the range of expected values (Figure 6b).

For MNTD, 75 countries (49.3%) showed significantly lower MNTD than expected, indicating strong phylogenetic clustering. Notable examples include Chile, China and Russia (Table S2). Again,

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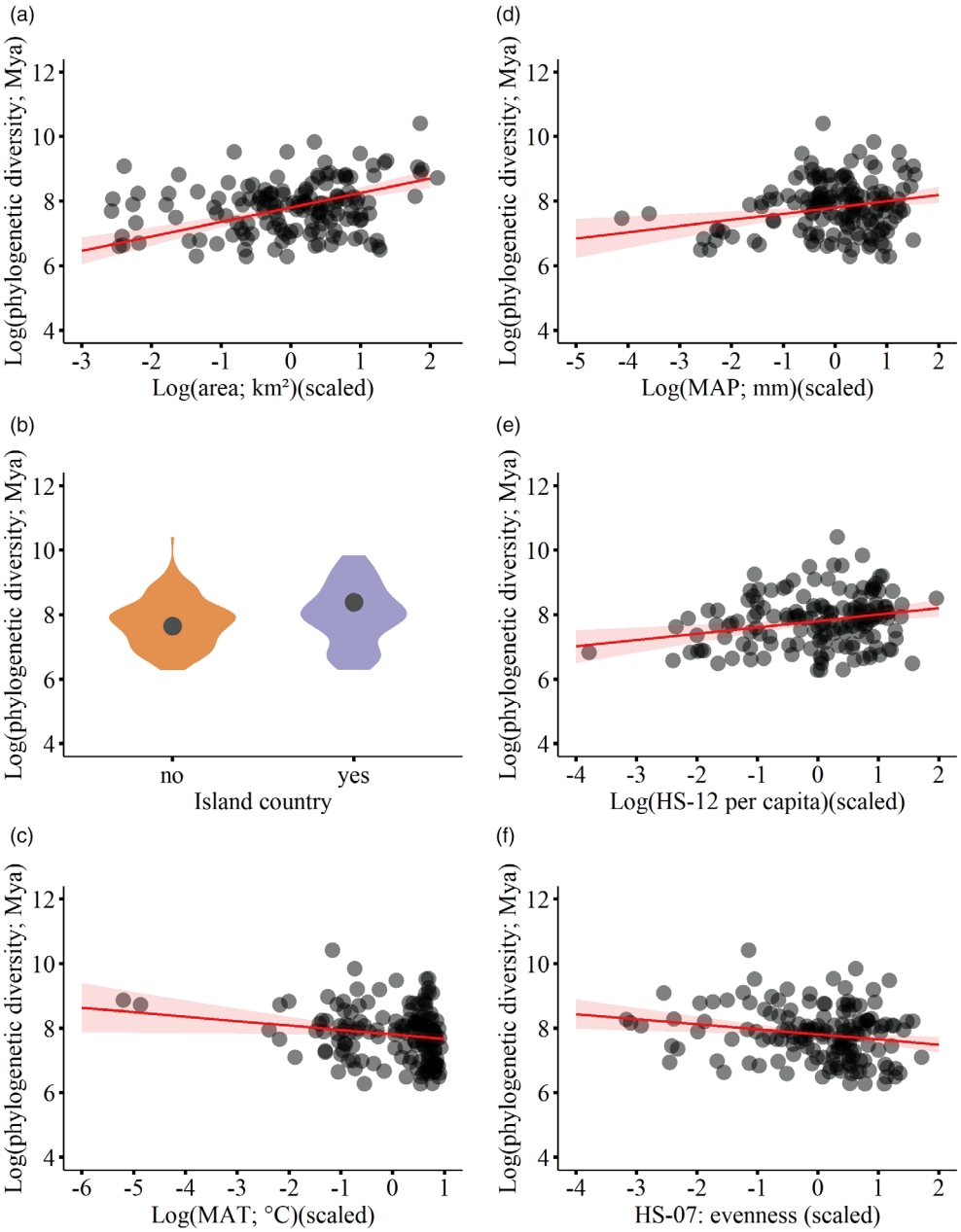

FIGURE 3 Phylogenetic diversity of invasive plants in 152 countries around the world, in relation to (a) land area, (b) insularity, (c) mean annual temperature (MAT), (d) mean annual precipitation (MAP), (e) the import values of the HS-12 category (seeds, grains and medicinal plants) per capita, and (f) evenness in import values of the HS-07 category (vegetables). The fitted lines with 95% confidence intervals from the minimum adequate LM model are shown.

only Mauritania exhibited a significantly higher MNTD than expected, suggesting phylogenetic overdispersion (Figure 6c). The remaining 78 countries (50.0%) had MNTD values within the range of expected values (Figure 6c).

4 | DISCUSSION

Information on factors determining invasive plant diversity is crucial for invasive plant management (Latombe et al., 2017; Pagad et al., 2015). We found that taxonomic and phylogenetic diversity, as well as the phylogenetic structure of invasive floras worldwide, varied considerably among countries. After accounting for variation in each country's land area, taxonomic and phylogenetic diversity were found to be closely correlated with geographic characteristics, climates, socio-economic status and international trade.

4.1 | Determinants of taxonomic diversity

The taxonomic diversity of invasive plants worldwide was, after correcting for the area of the country, influenced by multiple factors

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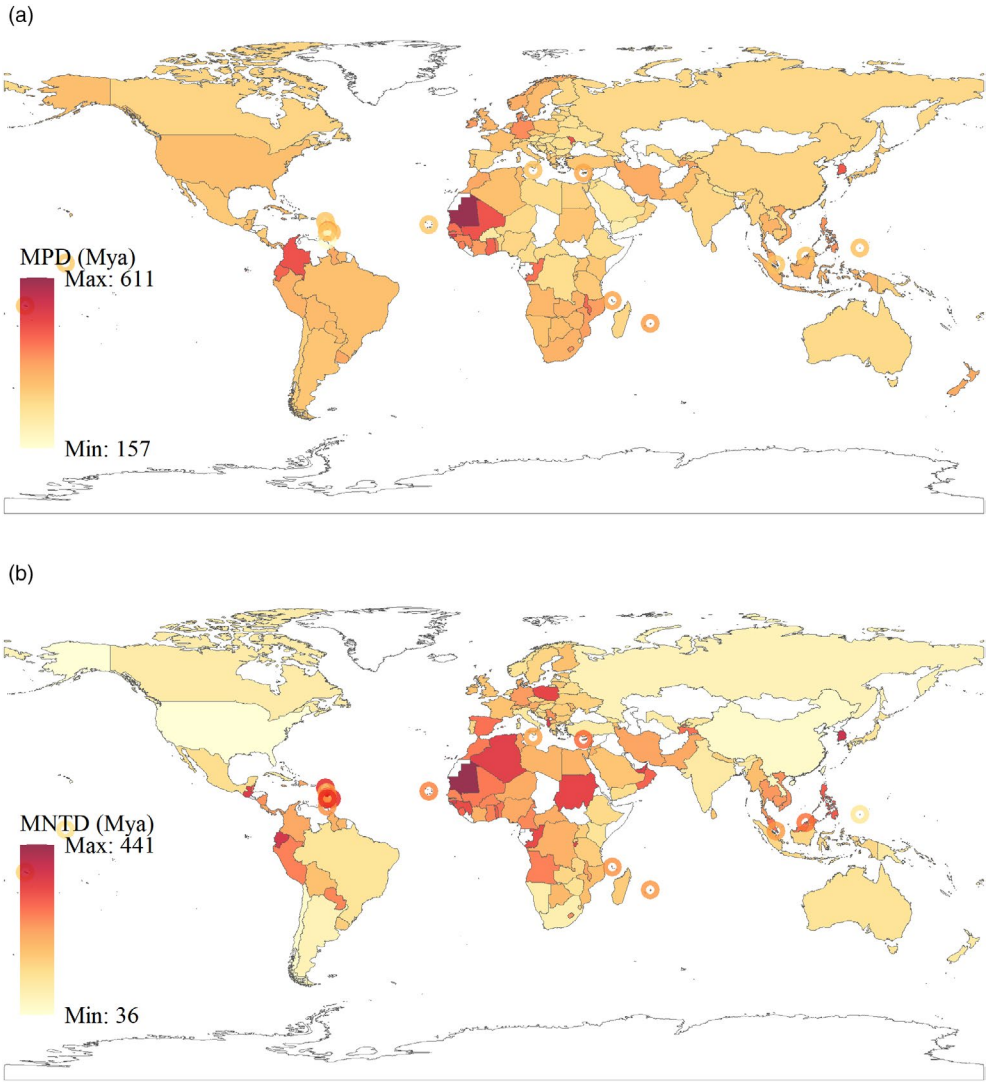

FIGURE 4 Global distribution of mean phylogenetic distance (MPD) and mean nearest taxon distance (MNTD) of invasive plants. Data on metrics of phylogenetic structure are available for 152 countries with more than one invasive species. Island countries smaller than 10,000 km^2 ($n = 16$) are marked with circles to ensure their visibility on the map. Regions with white filling have no data.

related to geographic characteristics, climates, socio-economic status and international trade. Insularity emerged as a strong predictor, and island countries exhibit a significantly higher diversity of invasive plants than mainland countries. Approximately 47.3% of the invasive plants have been recorded in the 33 island countries, even though the average area of these island countries is only one-seventh the size of mainland countries. This disproportionate concentration of invasive plants on islands aligns with previous studies, indicating that islands are more susceptible to plant invasions than mainland regions (Essl et al., 2019; Russell et al., 2017). One explanation for the high numbers of invasive species in island countries is the unique characteristics of native island floras, commonly referred to as the 'island plant syndrome' (Barton & Fortunel, 2023; Schrader et al., 2021). Native island floras are often thought to have evolved with conservative resource use, characterized by slow growth rates, limited dispersal ability and weak competitive abilities (Schrader et al., 2021; Westerband et al., 2021, 2024). These plant traits may lead to reduced biotic resistance of local native communities, thereby facilitating alien plant invasion (Barton & Fortunel, 2023; Schrader et al., 2021). Another plausible explanation is that the geographic characteristics of islands, including their remoteness, often result in unoccupied environmental niches (Essl et al., 2019; Moser et al., 2018). These insular conditions, coupled with recent anthropogenic disturbances, may promote the establishment success of alien plant species on islands (Bach et al., 2022; Moser et al., 2018). It should be noted that this observed pattern appears to challenge the island biogeography theory regarding reduced immigration on islands, but it implies that the few alien plants that did arrive through human-induced activity have a high probability of establishment. This elevated establishment success finally compensates for the low immigration rates, leading to significant invasions despite the islands' isolation (Moser et al., 2018).

Countries with high airport densities exhibited significantly elevated taxonomic diversity, highlighting the important role that

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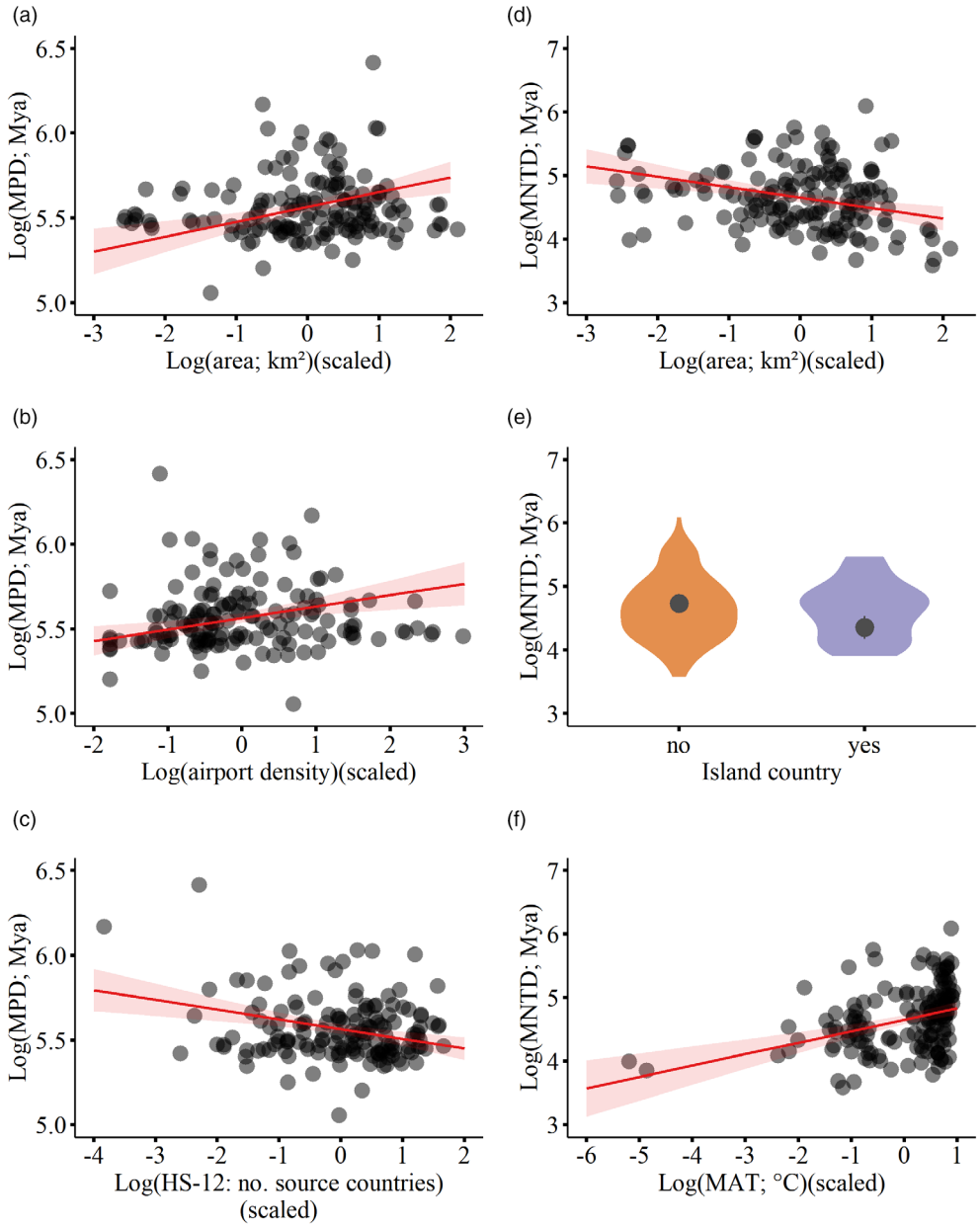

FIGURE 5 Mean phylogenetic distance (MPD) and mean nearest taxon distance (MNTD) of invasive plants in 152 countries around the world. MPD is in relation to (a) land area, (b) airport density and (c) the number of source countries that exported the HS-12 category (seeds, grains and medicinal plants); MNTD is in relation to (d) land area, (e) insularity and (f) mean annual temperature (MAT). The fitted lines with 95% confidence intervals from the minimum adequate LM model are shown.

transport infrastructures play in the introduction of invasive plant species across countries and continents (Dawson et al., 2017; Essl et al., 2019; Pouteau et al., 2021; Pyšek et al., 2010). This is because increased air traffic heightens the opportunities for accidental introductions through cargo shipments, passenger luggage and aircraft operations, thereby contributing to the spread of alien plant species beyond biogeographic barriers (Gallardo et al., 2015; Pouteau et al., 2021). However, the increased density of seaports did not result in a corresponding rise in taxonomic diversity, implying that airports serve as a more important pathway for invasive plant species. A possible reason for this discrepancy is that air transport, characterized by rapid transit times and higher turnover rates, may increase the likelihood that the propagules of invasive species are still viable upon arrival (Early et al., 2016).

The taxonomic diversity of invasive plants is also influenced by the regional climates, encompassing both MAT and MAP. Specifically, there was a negative correlation between taxonomic diversity and regional temperature, showing that invasive plants were more prevalent in warm temperate and (sub)tropical regions such as those in North America and East Asia, rather than in tropical areas like the Caribbean, Central Africa and South Asia (Essl et al., 2019). This is because relatively lower regional temperatures at higher latitudes may contribute to the persistence of low native plant diversity, thereby facilitating the spread of invasive plants with greater

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TABLE 2 Results of the minimum adequate linear models testing the effects of explanatory variables on mean phylogenetic distance (MPD) and mean nearest taxon distance (MNTD) of invasive plants in 152 countries worldwide.

Term	Estimate	SE	*t*-value	*p*-value	Ind. R^2	Ind. perc.
(a) MPD						
Intercept	5.56	0.01	391.31	<0.001	/	/
Airport density	0.07	0.02	3.21	0.002	0.022	19.4
Area	0.09	0.02	3.96	<0.001	0.042	36.61
HS-12: no. source countries	−0.06	0.02	−3.72	<0.001	0.050	43.99
(b) MNTD						
Intercept	4.74	0.04	111.18	<0.001	/	/
Insularity	−0.38	0.11	−3.50	0.001	0.025	12.73
Area	−0.16	0.04	−3.76	<0.001	0.055	27.87
MAT	0.18	0.04	4.87	<0.001	0.118	59.41

Note: The marginal R^2 of the model is 0.11 for MPD and 0.20 for MNTD. Individual R^2 values for each fixed factor and their percentage contribution to the overall R^2 are given. One trade category, HS-12 (seeds, grains and medicinal plants), was retained in the model of MPD.

diversity (Delavaux et al., 2023). In addition to regional temperature, regional precipitation plays a positive role in enhancing the taxonomic diversity of invasive plants. Adequate precipitation provides the necessary water resources that support the growth and expansion of a wide variety of invasive species (Ratcliffe et al., 2024). Correspondingly, regions with higher precipitation levels have more favourable habitats, allowing invasive plants to thrive and outcompete native vegetation. This is supported by several studies conducted in, for example, Asia (Inderjit et al., 2018; Wu et al., 2010), Europe (Lambdon et al., 2008) and Northern America (Stohlgren et al., 2005).

Interestingly, the taxonomic diversity of invasive plants is closely associated with the types of commodities originating from source countries. Some imported goods have a significant impact on the taxonomic diversity of invasive plants, while others have minimal impact. Based on existing evidence, it is clear that the imports of seeds, grains and medicinal plants (HS-12) and their evenness may greatly contribute to the introduction of invasive plants from diverse source countries, compared to the import of live plants and cut flowers (HS-06), vegetables (HS-07) and fruit and nuts (HS-08). A high HS-12 evenness also indicates that seeds, grains and medicinal plants are obtained in similar amounts from multiple regions, with no dominant source region. This diversity of origins likely increases the potential pool of species that could become invasive. Furthermore, the physical properties of seeds and propagules enable them to contaminate legitimate shipments and circumvent detection protocols during transit. In addition, many invasive plants have a history of being introduced for purposes such as medicines (e.g. *Datura stramonium* and *Phytolacca americana*) and have continued to spread in this fashion in many countries with traditional pharmacy practices like China (Weber et al., 2008; Zhang et al., 2022) and South Africa (Faulkner et al., 2020; Yessoufou et al., 2021). Consequently, the intentional or unintentional introduction and cultivation of alien plants inevitably increase propagule pressure and the risk of plant invasions (Guo et al., 2024).

Although some countries, like the United States, with relatively high import values for the HS-08 category (fruit and nuts) show high taxonomic diversity of invasive plants, the relationship between HS-08 and taxonomic diversity becomes negative after accounting for area. Similarly, the relationship between the taxonomic diversity of invasive plants and the evenness of the HS-07 category (vegetables) imports is negative. The absence of positive relationships may reflect stricter phytosanitary regulations for fruit, nut and vegetable imports. However, the reason for the negative relationship remains unclear. One possible explanation could be that fruit, nuts and vegetables are predominantly imported from warmer regions to colder regions, where the climate may be too cold for many accidentally introduced high-temperature adapted alien species to establish.

4.2 | Determinants of phylogenetic diversity and structure

As the evolutionary history covered by a group of species usually increases with the number of species in the group, it is not surprising that the spatial pattern in phylogenetic diversity of invasive plants aligns with the one for taxonomic diversity (Li et al., 2018; Park & Razafindratsima, 2019). Consequently, phylogenetic diversity increased with land area, MAP and import values of the HS-12 category per capita and was higher when the country is an island or archipelago instead of a mainland region. In contrast, phylogenetic diversity decreased with MAT and evenness in import values of the HS-07 category. This mirrored the trend observed for taxonomic diversity.

Our analyses of phylogenetic structure metrics (MPD and MNTD), which in contrast to phylogenetic diversity are not directly related to taxonomic diversity, revealed additional phylogenetic patterns among invasive plants worldwide. MPD exhibited positive correlations with land area and airport density. This suggests that countries with larger areas provide diverse habitats that

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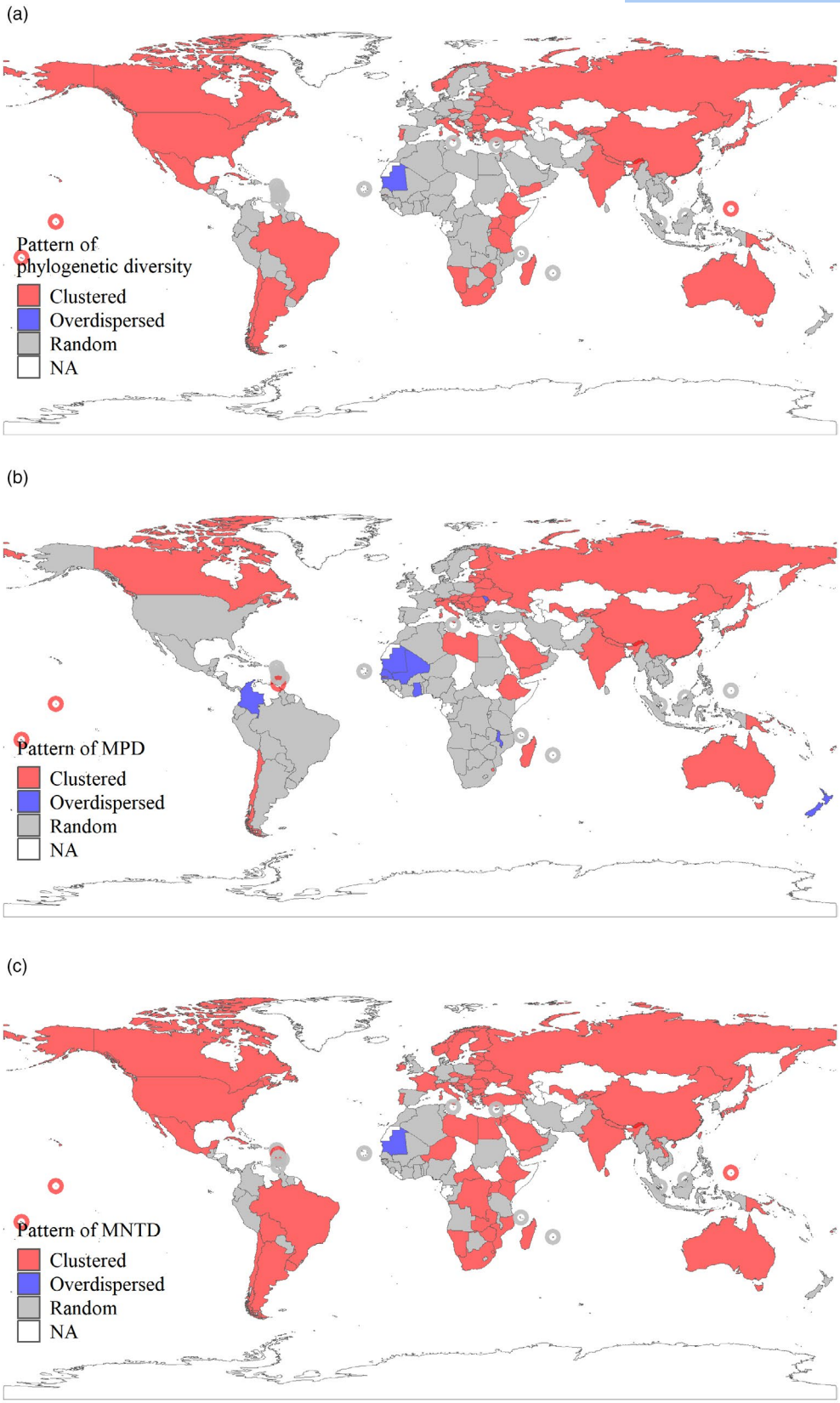

FIGURE 6 Global patterns of phylogenetic diversity (PD; a), mean phylogenetic distance (MPD; b) and mean nearest taxon distance (MNTD; c) of invasive plants. Data on phylogenetic diversity and structure are available for 152 countries with more than one invasive species. Red fillings indicate values of the phylogenetic metrics that are significantly lower than expected (i.e. phylogenetic clustering); blue fillings indicate values that are significantly higher than expected (i.e. phylogenetic overdispersion); and grey fillings indicate values that do not deviate from random expectations. Island countries smaller than 10,000 km^2 ($n = 16$) are marked with circles to ensure their visibility on the map. Regions with white filling have no data.

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support invasive species with diverse evolutionary histories and that transport infrastructures facilitate the long-distance introduction of alien plants from different source countries (Chapman et al., 2017; Hulme, 2021). Unexpectedly, MPD was negatively correlated with the number of source countries from which seeds, grains and medicinal plants were imported, implying that countries importing HS-12 category goods from fewer source countries tend to introduce invasive species from a wider range of phylogenetic clades. This phenomenon is particularly evident in countries in Sub-Saharan Africa (e.g. Mauritania), Latin America (e.g. Colombia) and the Caribbean. The exact reason for this pattern is unknown, but, to the best of our knowledge, this study is the first to reveal that international trade serves as a key mechanism shaping the global pattern of MPD.

Compared to MPD, the global distribution pattern of MNTD is predominantly shaped by the climate, with higher values of MNTD observed in mainland countries that are small and/or have warm climates. This pattern is evident across Africa (e.g. Mauritania, Algeria and Sudan), Latin America and the Caribbean (e.g. Guatemala and Ecuador) and Asia (e.g. United Arab Emirates). One plausible explanation for this trend is that these regions tend to have relatively few invasive plant species, thereby reducing the likelihood that any two species are closely related. Overall, our findings reveal two distinct mechanisms that shape the phylogenetic patterns of invasive plant species worldwide: Human activities drive broader phylogenetic diversity, while environmental filtering through climatic conditions determines fine-scale phylogenetic structure.

4.3 | Regional clustering of phylogenetic diversity

Using randomization tests of phylogenetic diversity measures, we quantified global patterns of phylogenetic clustering in invasive plants. Our analysis revealed that 28.9%–49.3% of studied countries exhibited phylogenetic clustering, while only 0.6%–5.3% showed phylogenetic overdispersion. These findings indicate that invasive plants within many countries tend to be more closely related to one another than expected by chance (e.g. China in Qian et al., 2022; Canada and the United States in Qian & Sandel, 2023).

In addition, the prevalence of phylogenetic clustering also showed distinct geographical patterns among continents (see Figure S2 for more details). A higher proportion of countries in North America, Oceania and Europe consistently exhibited phylogenetic clustering of invasive plants across all three metrics, while a lower proportion in Asia, Africa and South America exhibited phylogenetic clustering. In particular, Africa was the only continent that consistently showed evidence of phylogenetic overdispersion. To some extent, such a phylogenetic clustering pattern likely reflects regional differences in introduction histories: Developed countries may experience repeated introductions of closely related alien plant groups through established trade networks, while developing countries receive more diverse plant species through varied and less systematic pathways.

5 | CONCLUSIONS

The global distribution of invasive plant diversity is fundamentally driven by the interplay of geographic characteristics, climates, socio-economic status and international trade. Island countries harbour disproportionately high invasive plant diversity due to their distinct evolutionary trajectories and inherent ecological vulnerability. Transport infrastructure, particularly airports, emerges as an important entry point for invasive plant species, and international trade—specifically in seeds, grains and medicinal plants—substantially influences their global distribution. Regional climate regulates the establishment of invasive plants, and the highest diversity was found in countries with lower temperatures (associated with higher latitude) and elevated precipitation levels. In addition, developed regions (e.g. North America, Oceania and Europe) exhibit a higher proportion of countries with strong phylogenetic clustering of invasive plants, possibly resulting from repeated introductions of related species through established trade networks, while developing regions manifest more diverse phylogenetic patterns reflecting various introduction pathways. These findings highlight the importance of integrating both taxonomic and phylogenetic perspectives in invasion ecology, emphasizing the need for regionally tailored management strategies that effectively account for regional geographic, climatic, socio-economic and trade-related factors to mitigate future plant invasions.

AUTHOR CONTRIBUTIONS

Bi-Cheng Dong, Lan-Hui Wang and Fei-Hai Yu conceived and designed research; Bi-Cheng Dong and Li-Yuan Gao performed research; Bi-Cheng Dong, Lan-Hui Wang, and Li-Yuan Gao contributed data; Bi-Cheng Dong and Li-Yuan Gao analysed data with the support of Mark van Kleunen and Fei-Hai Yu; Bi-Cheng Dong, Lan-Hui Wang and Li-Yuan Gao drafted the paper, and all authors have revised the subsequent drafts. All authors have contributed substantially to the interpretation and discussion of the results.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

The data and R codes that support the findings of this study are openly available in Dryad Digital Repository at https://doi.org/10.5061/dryad.h18931zxz (Dong et al., 2025).

ORCID

Bi-Cheng Dong https://orcid.org/0000-0002-9658-9969
Mark van Kleunen https://orcid.org/0000-0002-2861-3701
Fei-Hai Yu https://orcid.org/0000-0001-5007-1745

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1: Detailed description of the four HS-2 commodity groups associated with plant-derived products.

Table S2: Results of randomization tests for phylogenetic clustering patterns in top-ranked countries.

Figure S1: A heatmap of the correlation matrix between transformed geographic, climatic, socio-economic, and international trade variables of 152 countries: airport density, seaport density, GDP per capita, land area, human-population density, mean annual temperature (MAT), mean annual precipitation (MAP), import values of HS categories per capita [i.e., HS-06 (live plants and cut flowers), HS-07 (vegetables), HS-08 (fruit and nuts), and HS-12 (seeds, grains, and medicinal plants)], the number of source countries that exported each HS category, and the evenness in import values of each HS category across countries.

Figure S2: Stacked bar plots illustrating the proportions of three phylogenetic metrics across different continental scales and at the island level.

Figure S3: Relationships between native species richness and taxonomic diversity and phylogenetic diversity of invasive species across 152 countries.

Figure S4: Relationship between native species richness and land area across 152 countries.

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